Systemic Risk Exchange (SRX)

A Framework for Real-Time Systemic Risk Measurement,
Walk-Forward Crisis Detection, Portfolio-Specific Crash Protection Pricing,

and Clearinghouse Resilience Simulation

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Abstract

We present the Systemic Risk Exchange (SRX), a research prototype that integrates real-time systemic stress detection, portfolio-specific crash protection pricing, and clearinghouse-grade capital adequacy modeling into a single analytical framework.

The platform introduces two composite indices. The **Systemic Risk Score (SRS)** is a 0–100 measure of real-time market stress derived from four cross-asset signals: volatility (V_t), Amihud illiquidity (L_t), pairwise correlation (C_t), and drawdown acceleration (M_t). The **Global Systemic Risk Index (GSRI)** aggregates asset-class-specific stress across equities, credit, duration, digital assets, and cross-asset correlation into an institutional benchmark.

A novel **Shock Multiplier** mechanism amplifies the SRS and GSRI by 1.25× when volatility or drawdown scores double within five trading days, ensuring the framework detects both slow structural decay and sudden exogenous dislocations.

We validate the framework against three historical crises using a strict walk-forward protocol with zero look-ahead bias. The GSRI provided a **144-day structural early warning** before the 2008 Lehman collapse and detected the 2020 COVID-19 crash within **5 trading days** via the Shock Multiplier. The 2022 rate-tightening cycle was classified as an Early Warning regime with sustained elevated readings throughout H1 2022.

The crash protection pricing engine produces actuarial premium estimates that scale quadratically with systemic stress. A five-layer default waterfall and SRS-driven dynamic circuit breaker system complete the clearinghouse simulation. All components are implemented as a working research prototype with an 8-page interactive dashboard.

Keywords: systemic risk, crisis detection, crash protection, walk-forward validation, portfolio hedging, clearinghouse design, financial infrastructure

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Introduction

Systemic financial crises remain among the most consequential and least efficiently hedged risks in global capital markets. The 2008 Global Financial Crisis produced aggregate losses exceeding $2 trillion in the US banking sector. The 2020 COVID-19 crash erased 34% of S&P 500 market capitalization in 23 trading days. The 2022 rate-tightening cycle triggered simultaneous drawdowns across equities, fixed income, and digital assets—violating the diversification assumptions underlying most institutional portfolios.

Despite the magnitude of these events, the available hedging infrastructure suffers from three structural deficiencies:

1. **Basis risk.** Portfolio managers hedge with index-level instruments (S&P 500 puts, VIX derivatives) that bear limited resemblance to their actual multi-asset exposures. A portfolio with 40% equities, 20% high-yield credit, 20% duration, 10% commodities, and 10% digital assets has a fundamentally different risk profile than the S&P 500 alone.

2. **Procyclical cost.** Protection becomes most expensive precisely when it is most needed. VIX call spreads that cost 50 basis points in calm markets can exceed 400 basis points during stress—pricing out the institutions that need protection most urgently.

3. **Fragmented analytics.** Systemic risk measurement (academic research, central bank monitoring) and crash protection markets (options exchanges, OTC derivatives) operate as disconnected systems. No existing infrastructure links real-time systemic stress detection to dynamic, portfolio-specific protection pricing.

SRX addresses these gaps by proposing an integrated framework that: (a) measures systemic stress in real-time from cross-asset market signals; (b) translates stress into portfolio-specific risk scores; (c) prices crash protection dynamically as a function of systemic conditions; and (d) simulates clearinghouse resilience under extreme scenarios.

This paper describes the mathematical methodology behind each component, validates the crisis detection capabilities against three historical regimes, and outlines a staged development path from research prototype to potential institutional infrastructure.

System Architecture

The SRX platform consists of five analytical layers, each consuming the output of the layer below:

1. **Market Data Layer** — ingests and caches daily price, volume, and return data across six core asset classes (SPY, TLT, HYG, GLD, UUP, BTC-USD) plus a configurable extended ticker universe.

2. **Portfolio Risk Index (PRI)** — decomposes user-submitted portfolios into four standardized risk dimensions and constructs a normalized portfolio time series.

3. **Systemic Risk Engine (SRS / GSRI)** — computes real-time systemic stress from four cross-asset signals with non-linear scoring and shock amplification.

4. **Crash Protection Pricing Engine** — estimates actuarial premiums calibrated to systemic conditions, portfolio risk, and protection parameters.
5. **Clearinghouse Simulation Layer** — models a five-layer default waterfall and SRS-driven dynamic circuit breakers.

`Market Data` → `PRI Engine` → `SRS/GSRI Engine` → `Pricing Engine` → `Clearinghouse`

Each layer is walk-forward: computations at time t use only data from $[0, t]$. No future information leaks between layers.

Figure 1: SRX five-layer pipeline. Downstream layers inherit the walk-forward constraint.

The current implementation is a monolithic Python application (approximately 8,000 lines across 9 modules) using Streamlit for interactive visualization. The architecture is designed for eventual decomposition into independent microservices.

Portfolio Risk Index (PRI)

Portfolio Specification

Participants submit portfolios as weighted baskets of tradeable instruments:

$$\mathcal{P} = \{(w_1, \tau_1), (w_2, \tau_2), \ldots, (w_N, \tau_N)\}, \quad \sum_{i=1}^{N} w_i = 1, \quad w_i > 0$$

where w_i is the portfolio weight and τ_i is the ticker symbol for asset i. The platform supports equities, fixed income ETFs, commodities, and digital assets.

PRI Time Series

The Portfolio Risk Index is constructed as a normalized cumulative return series beginning at 100:

$$PRI_t = 100 \times \prod_{s=1}^{t} \left(1 + \sum_{i=1}^{N} w_i \cdot r_{i,s}\right)$$

where $r_{i,s}$ is the daily return of asset i on day s. This construction preserves the compounding dynamics of the actual portfolio and enables direct comparison across different portfolio compositions.

Composite Risk Score

The PRI composite score (0–100) aggregates four risk dimensions:

$$PRI_{score} = \alpha_V \cdot S_V + \alpha_C \cdot S_C + \alpha_K \cdot S_K + \alpha_T \cdot S_T \tag{1}$$

Component	Measures	Symbol	Mechanism
Volatility	Price dispersion	S_V	Rolling annualized σ of portfolio returns
Correlation	Diversification loss	S_C	Average pairwise ρ among portfolio assets
Concentration	Weight imbalance	S_K	Herfindahl-Hirschman Index of w_i
Tail Risk	Distribution shape	S_T	Skewness and excess kurtosis of return distribution

Table 1: PRI composite score components. Each sub-score is mapped to $[0, 100]$ via the non-linear scoring function defined in Section 4.3.

Systemic Risk Engine

Design Principles

The Systemic Risk Score (SRS) is a composite 0–100 index measuring real-time systemic market stress. Three design principles govern its construction:

1. **Walk-forward computation.** For any date t, all signals are computed using only data from the rolling window $[t - w, t]$. No future data is visible.
2. **Non-linear sensitivity.** The scoring function amplifies the distinction between "stressed" and "crisis" relative to the distinction between "calm" and "slightly stressed," reflecting the convex dynamics of systemic crises.
3. **Shock amplification.** A multiplicative accelerator fires when sub-signals double within five days, capturing sudden dislocations that longer lookback windows would otherwise dilute.

Signal Definitions

Volatility Signal (V_t)

Rolling annualized volatility averaged across all tracked assets:

$$V_t = \frac{1}{N} \sum_{i=1}^{N} \left(\sqrt{252} \cdot \sqrt{\frac{1}{w-1} \sum_{s=t-w}^{t} (r_{i,s} - \bar{r}_i)^2} \right)$$

where w is the lookback window (default: 20 trading days) and $\bar{r}_i$ is the mean return of asset i over the window. During systemic events, V_t rises as cross-asset price dispersion increases simultaneously.

Liquidity Signal (L_t) — Amihud Illiquidity Ratio

Following Amihud (2002), the illiquidity signal captures price impact per unit of trading volume:

$$L_t = \frac{10^9}{N} \sum_{i=1}^{N} \frac{1}{w} \sum_{s=t-w}^{t} \frac{|r_{i,s}|}{V_{i,s}^{\$}}$$

where $r_{i,s}$ is the daily return and $V^{\$}_{i,s}$ is the dollar volume for asset i on day s. The 10^9 scaling factor converts the raw ratio to a human-readable scale (typical values: 0.1 in calm markets, 20–50 during crises).

During systemic stress, price movements become disproportionately large relative to available volume as market makers widen spreads and liquidity providers withdraw. This makes L_t a leading indicator of market fragility.

Correlation Signal (C_t)

Rolling average off-diagonal pairwise correlation:

$$C_t = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j>i} \rho^{(w)}_{ij,t}$$

where $\rho^{(w)}_{ij,t}$ is the rolling w-day Pearson correlation between the returns of asset i and asset j.

When C_t approaches 1.0, all assets move in lockstep—diversification has collapsed, and systemic contagion is propagating across asset classes. This signal was the primary driver of the 2008 GFC early warning, as credit, equity, and commodity correlations rose steadily for months before the Lehman collapse.

Drawdown Signal (M_t)

Drawdown acceleration measures how rapidly prices are falling from their rolling peaks:

$$M_t = \frac{1}{N} \sum_{i=1}^{N} \left| \frac{P_{i,t} - P^{peak}_{i,t}}{P^{peak}_{i,t}} \right|, \qquad P^{peak}_{i,t} = \max_{s \in [t-w,t]} P_{i,s}$$

where $P_{i,t}$ is the price and $P^{peak}_{i,t}$ is the rolling peak. Unlike cumulative drawdown (which can remain elevated during slow recoveries), this measure resets with the rolling window, making it sensitive to *acceleration* in price declines.

Non-Linear Scoring Function

Each raw signal value x is mapped to a score $\phi(x) \in [0, 100]$ via:

$$\phi(x; x_{low}, x_{high}, \gamma) = 100 \times \left(\frac{\text{clamp}(x,\ x_{low},\ x_{high}) - x_{low}}{x_{high} - x_{low}} \right)^{\gamma}$$

where $\text{clamp}(x, a, b) = \min(b, \max(a, x))$ and $\gamma > 1$ is the convexity parameter. The power curve ensures that the score *accelerates* as the raw signal approaches crisis levels:

- At $\gamma = 1.0$ (linear), a signal at 50% of its crisis range scores 50.
- At $\gamma = 1.5$ (convex), the same signal scores only $50^{1.5}/100 \approx 35.4$—requiring more extreme conditions to reach high scores.

This matches the empirical observation that systemic crises develop non-linearly: slowly at first (rising correlations, thinning liquidity), then very fast (volatility explosion, forced selling, feedback loops).

Signal	**Weight** (λ)	x_{low} **(Calm)**	x_{high} **(Crisis)**	γ
V_t (Volatility)	0.30	5% ann.	60% ann.	1.5
L_t (Liquidity)	0.20	0	50 (scaled)	1.3
C_t (Correlation)	0.25	-0.2	$+0.8$	1.4
M_t (Drawdown)	0.25	0%	5% accel.	1.6

Table 2: SRS signal weights and calibration thresholds. Thresholds are set a priori from historical crisis calibration and are not adjusted per-period.

SRS Aggregation

The four scored sub-signals are combined via calibrated weights:

$$SRS_t = \lambda_V \cdot \phi(V_t) + \lambda_L \cdot \phi(L_t) + \lambda_C \cdot \phi(C_t) + \lambda_M \cdot \phi(M_t) \tag{2}$$

The result is clamped to $[0, 100]$ after the Shock Multiplier is applied.

Shock Multiplier

Standard rolling-window indicators dilute sudden dislocations: a 60-day lookback that was calm for 55 days and violent for 5 days produces a moderate average. To address this, we introduce a multiplicative acceleration layer:

$$SRS_t^{adj} = \min\Big(100,\ SRS_t \times \xi_t\Big) \tag{3}$$

where the Shock Multiplier ξ_t is defined as:

$$\xi_t = \begin{cases} 1.25 & \text{if } \dfrac{\phi(V_t)}{\phi(V_{t-5})} \geq 2.0 \text{ or } \dfrac{\phi(M_t)}{\phi(M_{t-5})} \geq 2.0, \text{ and } \phi(V_{t-5}) > 5 \\ 1.00 & \text{otherwise} \end{cases}$$

The threshold $\phi(V_{t-5}) > 5$ prevents the multiplier from firing on negligible signals (e.g., volatility rising from 0.1 to 0.3). The 5-day window corresponds to one trading week—the timescale of institutional forced-selling cascades.

This mechanism is critical for detecting exogenous shocks (COVID-19, flash crashes) where the transition from calm to crisis occurs in under one week.

Global Systemic Risk Index (GSRI)

Construction

While the SRS measures aggregate cross-asset stress, the GSRI provides a decomposed view by asset class:

$$GSRI_t = \sum_{k=1}^{K} \omega_k \cdot \phi(S_{k,t}) \times \xi_t^{GSRI} \tag{4}$$

where ξ_t^{GSRI} is the Shock Multiplier applied to the GSRI using equity volatility and cross-asset correlation as acceleration signals.

Component	Market Proxy	Weight (ω_k)	Scoring
Equity Volatility	SPY rolling vol	0.25	$\phi(x; 0.05, 0.60, 1.5)$
Credit Stress	HYG vol + HYG-TLT spread	0.25	Blended (60/40)
Treasury Volatility	TLT rolling vol	0.15	$\phi(x; 0.03, 0.30, 1.3)$
Crypto Volatility	BTC-USD rolling vol	0.10	$\phi(x; 0.20, 1.20, 1.2)$
Cross-Asset Correlation	Average pairwise ρ	0.25	$\phi(x; -0.2, 0.8, 1.4)$

Table 3: GSRI component weights and scoring parameters.

The credit stress component uses a blended signal: 60% HYG volatility, 40% HYG-TLT performance spread. When high-yield bonds underperform Treasuries, credit risk is repricing—a key leading indicator of systemic stress.

Risk Classification

Range	Zone	Classification	Institutional Implication
0–29	Normal	Standard conditions	Routine monitoring
30–49	Monitoring	Emerging stress	Hedging review recommended
50–74	Elevated	2σ event; hedging entry zone	Portfolio protection warranted
75–100	Critical	3σ event; systemic failure	Emergency risk controls

Table 4: GSRI risk classification thresholds with institutional interpretation.

Historical Crisis Validation

Walk-Forward Protocol

Validation uses a strict out-of-sample methodology. For each trading day t in a historical crisis window:

1. Data is sliced to $[t - w, t]$ where w is the lookback window.
2. All four signals (V_t, L_t, C_t, M_t) are computed from this slice only.
3. The non-linear scoring function and Shock Multiplier are applied.
4. The composite GSRI is recorded.

No future data is visible at any step. The scoring thresholds (Table 2) are fixed a priori and are **not** re-calibrated per crisis—the same parameters are used across all three validation periods.

Adaptive Sensitivity

The lookback window w governs the sensitivity–stability tradeoff. Shorter windows use tighter scoring thresholds to detect rapid dislocations; longer windows use deeper thresholds to require sustained stress:

Window	Vol Range	Corr Range	DD Range	Detection Profile
20d	6%–28%	0.05–0.55	0%–6%	Shock: flash crashes, sudden dislocations
40d	7%–32%	0.03–0.60	0%–7%	Fast: weekly regime shifts
60d	8%–38%	0.00–0.65	0%–9%	Balanced (default)
90d	9%–42%	0.00–0.70	0%–11%	Structural: sustained stress
120d	10%–48%	0.00–0.75	0%–13%	Deep: chronic systemic rot

Table 5: Adaptive sensitivity thresholds by lookback window. These apply to the historical validation engine only; the live GSRI uses the fixed thresholds in Table 2.

Regime Classification

The validation engine automatically classifies each crisis by the pattern of GSRI behavior:

- **Structural Decay:** GSRI $\geq$ 50 more than 100 trading days before the crash date. Indicates slow, sustained buildup of systemic fragility—rising correlations, deteriorating liquidity, increasing concentration risk accumulating over months.
- **Exogenous Shock:** GSRI $\geq$ 50 within 10 trading days of the crash *and* the Shock Multiplier was activated. Indicates a sudden, external dislocation that bypassed the gradual stress buildup pathway.
- **Early Warning:** GSRI $\geq$ 50 between 10 and 100 days before the crash. Provides actionable hedging lead time.

Validated Results

2008 Global Financial Crisis

Using the default 60-day lookback and the SRX historical ticker universe (SPY, TLT, HYG, GLD, EFA, EEM, QQQ, IWM):

- GSRI first crossed Elevated (50) approximately March 2008
- Lehman Brothers collapse: September 15, 2008
- **Lead time: ∼144 trading days**
- **Regime: Structural Decay**
- Primary driver: rising cross-asset correlation (C_t) and credit spread widening, consistent with the subprime contagion narrative
- The GSRI remained above Elevated for the duration of the crisis, peaking above 90 in Q4 2008

2020 COVID-19 Crash

- WHO pandemic declaration: March 11, 2020
- Shock Multiplier activated within 5 trading days as equity volatility doubled
- **Lead time: 5 trading days (Shock Multiplier-assisted)**
- **Regime: Exogenous Shock**
- Primary driver: extreme volatility acceleration (V_t doubling in <5 days)
- The 60-day lookback alone produced a pre-crash GSRI of ∼38; the Shock Multiplier elevated this to 47+, crossing into Monitoring range

2022 Rate Tightening Cycle

- GSRI detected sustained elevated stress throughout H1 2022
- Signal reflected simultaneous equity, fixed income, and digital asset drawdowns—a regime where traditional 60/40 diversification failed
- **Regime: Early Warning (mixed structural/exogenous)**
- The correlation signal (C_t) was the primary driver as equities and bonds fell together

The validation demonstrates that the *same* framework, *without parameter re-tuning*, detects both slow-building structural crises and sudden exogenous shocks—the two fundamental failure modes of systemic risk.

Crash Protection Pricing Engine

Premium Formula

The pricing engine estimates crash protection premiums using a multiplicative actuarial framework:

$$\Pi = N \times \beta \times \mu_{SRS} \times \mu_{PRI} \times \mu_L \times \kappa \times \delta \times \eta \tag{5}$$

Symbol	Component	Typical Range
N	Notional exposure	User-defined
β	Base rate	0.01 (1% of notional)
μ_{SRS}	Systemic risk multiplier	1.0–4.0×
μ_{PRI}	Portfolio risk adjustment	0.5–3.0×
μ_L	Liquidity adjustment	0.8–2.0×
κ	Risk loading factor	1.25×
δ	Duration factor: $\sqrt{d/90}$	0.6–2.0×
η	Depth factor: $\sqrt{20/p}$	0.6–2.0×

Table 6: Crash protection premium components. d = contract duration in days, p = protection level in percent.

SRS-Driven Convex Pricing

The SRS multiplier exhibits quadratic behavior above the Elevated threshold:

$$\mu_{SRS} = \begin{cases} 1.0 & \text{if } SRS_t \leq 50 \\ \left(\dfrac{SRS_t}{50}\right)^2 & \text{if } SRS_t > 50 \end{cases} \tag{6}$$

At $SRS = 80$, the multiplier is 2.56×; at $SRS = 100$, it reaches 4.0×. This convexity serves two purposes: (a) it reflects the genuine increase in expected loss as systemic stress rises, and (b) it moderates speculative demand during crises, preserving capacity for genuine hedging.

Clearinghouse Framework

Five-Layer Default Waterfall

Loss absorption follows a structured capital hierarchy inspired by CPMI-IOSCO Principles for Financial Market Infrastructures (2012):

Layer	Description	Default Size	Absorbs First
1	Defaulter Initial Margin	$200M	✓
2	Defaulter Default Fund Contribution	$50M	
3	Defaulter Contingent Capital	$30M	
4	Mutualized Default Fund	$500M	
5	SRX Clearinghouse Capital	$150M	
	Total Waterfall Capacity	**$930M**	

Table 7: Default waterfall layer structure. Losses flow sequentially through layers 1–5.

Each layer reports its status as INTACT, PARTIALLY DEPLETED, or EXHAUSTED. The simulation accepts configurable layer sizes and loss magnitudes, enabling stress testing of waterfall adequacy under various scenarios.

Dynamic Circuit Breakers (Gating)

The platform implements SRS-driven automatic risk controls that activate continuously, preventing the discrete jumps that can trigger procyclical forced selling:

Gate	SRS Range	Automatic Actions
0 — Normal	≤ 70	Standard operation. Base margin requirements.
1 — Elevated	70–85	Margin multiplier 1.25×+. New contracts permitted with enhanced monitoring.
2 — Restricted	85–95	Margin multiplier 1.50×+. New contract origination suspended.
3 — Critical	> 95	Margin multiplier 2.0×+. Contingent capital activated. Redemptions delayed. Forced deleveraging authorized.

Table 8: Dynamic gating thresholds and automatic risk control actions.

Margin multipliers increase continuously (piecewise linear interpolation) between gate boundaries, smoothing the transition and reducing the risk of cliff effects.

Stress Testing

Five preset scenarios apply asset-class-specific shocks calibrated to historical crisis magnitudes:

Scenario	Equity	Credit	Duration	Crypto	Vol ×
2008 Financial Crisis	−38%	−30%	+8%	−50%	3.5×
COVID-19 Crash	−34%	−15%	+5%	−40%	4.0×
Interest Rate Shock	−18%	−20%	−12%	−25%	2.0×
Liquidity Freeze	−22%	−35%	+3%	−45%	3.0×
Crypto Contagion	−12%	−10%	+2%	−65%	2.5×

Table 9: Stress scenario shock parameters. Duration assets (TLT) appreciate in flight-to-quality scenarios and depreciate in rate shocks.

For each scenario, the engine computes: projected portfolio drawdown, stressed PRI and SRS, per-asset loss attribution, worst contributing asset, and estimated recovery time in trading days.

Implementation Details

Component	Technology
Analytical engines	Python 3.12+ (NumPy, pandas)
Data source	Yahoo Finance via `yfinance` (`curl_cffi` for TLS fingerprinting)
Interactive charts	Plotly with range selectors, risk zone shading, shock markers
Dashboard	Streamlit (8 pages, monolithic deployment)
Hosting	Streamlit Community Cloud

Table 10: Implementation technology stack.

Module inventory:

- `data/market_data.py` — data ingestion, caching, column normalization
- `backend/portfolio_engine.py` — PRI computation and portfolio decomposition
- `backend/gsri_engine.py` — SRS and GSRI with Shock Multiplier
- `backend/pricing_engine.py` — crash protection premium estimation
- `backend/stress_engine.py` — five-scenario stress testing
- `backend/clearinghouse_simulation.py` — default waterfall simulation
- `backend/gating_engine.py` — dynamic circuit breaker logic
- `backend/historical_validation.py` — walk-forward crisis backtesting with adaptive sensitivity and regime classification
- `frontend/dashboard.py` — 8-page interactive Streamlit dashboard with scrollable GSRI history, shock markers, and risk zone visualization

Development Roadmap

Phase	Stage	Description
1	Research Prototype	Working software demonstrating all analytical components. Walk-forward validation against three historical crises. *(Current stage.)*
2	Public Analytics	Deploy GSRI as a publicly accessible systemic risk benchmark. API access for institutional subscribers.
3	Institutional Pilots	Partner with hedge funds and asset managers for portfolio-specific risk analytics and custom ticker universes.
4	Bilateral Contracts	Enable OTC crash protection contracts priced by the SRX engine between institutional counterparties.
5	Market Infrastructure	Potential evolution into a regulated exchange or clearinghouse for systemic risk transfer.

Table 11: Staged development roadmap. Each phase is designed to be viable as a standalone business.

Conclusion

Systemic risk remains one of the least efficiently transferred risks in global financial markets. The instruments available for hedging systemic crashes are indirect, procyclically expensive, and structurally misaligned with the portfolios they are meant to protect.

SRX demonstrates that it is technically feasible to integrate real-time systemic stress measurement, portfolio-specific risk scoring, dynamic crash protection pricing, and clearinghouse-grade capital adequacy modeling into a single analytical framework. The walk-forward validation against three distinct crisis regimes—2008 structural decay, 2020 exogenous shock, 2022 mixed regime—*without parameter re-tuning* provides evidence that the underlying methodology captures both fundamental modes of systemic failure.

The Shock Multiplier mechanism represents a novel contribution: by detecting intra-week acceleration in volatility or drawdown scores, the framework responds to sudden dislocations that standard rolling-window indicators dilute. This is the mechanism that makes the 2020 COVID detection possible within 5 trading days.

While currently implemented as a research prototype, the architecture is designed for staged evolution toward institutional financial infrastructure. The immediate deployment path is a public systemic risk analytics platform; the longer-term vision is a marketplace for portfolio-specific crash protection.

Disclaimer. SRX is a research prototype developed for analytical and educational purposes. It is not a regulated exchange, trading system, or financial product. Nothing in this paper or the associated software constitutes financial advice, an offer to trade, or a solicitation of any kind. All calculations are model-based estimates subject to significant limitations including data availability, model specification risk, and calibration uncertainty. Historical validation results reflect the behavior of specific indices over specific periods and should not be interpreted as predictive of future performance.

References

[1] Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.

[2] CPMI-IOSCO (2012). Principles for Financial Market Infrastructures. Bank for International Settlements.

[3] Brownlees, C. and Engle, R.F. (2017). SRISK: A Conditional Capital Shortfall Measure of Systemic Risk. *Review of Financial Studies*, 30(1), 48–79.

[4] Acharya, V.V., Pedersen, L.H., Philippon, T., and Richardson, M. (2017). Measuring Systemic Risk. *Review of Financial Studies*, 30(1), 2–47.

[5] Adrian, T. and Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.

[6] Bisias, D., Flood, M., Lo, A.W., and Valavanis, S. (2012). A Survey of Systemic Risk Analytics. *Annual Review of Financial Economics*, 4, 255–296.

[7] Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.